

Toward a Behavioral Science of Human-AI Teaming

Diagnosing Collaboration Health as a Dynamic Relationship

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Abstract

Human-AI collaboration is a *dynamic relationship*—one that forms, deepens, plateaus, and dissolves along observable behavioral trajectories. My position is that the teaming construct, as currently used, is aimed at the wrong unit of analysis to diagnose it. Team cognition, including its explicitly dynamic strands, theorizes the emergence of *collective* states among members who reciprocate: shared mental models, transactive memory, closed-loop coordination. Yet the two trajectories that most often decide enterprise AI deployments are neither collective nor reciprocal—they are the individual practice of one worker addressed to a partner that does not reciprocate: whether that worker is *deepening* their practice, and whether they are quietly *disengaging*. Those belong to expertise-development and technology-discontinuance theory, not to team cognition. To make this concrete I put a candidate instrument on the table: the **HAI Behavioral Lens**, four telemetry-observable behavioral axes—delegation, steering, abandonment, and deepening. Two map cleanly onto teaming vocabulary; two fall outside it, and they are the two that most often decide whether a deployment holds. That asymmetry has a methodological consequence: the right first question is not “are they a team?” but “is this person working well with AI at all?”—and the right output is not a score but a *collaboration health profile* read across the axes over time.

CCS Concepts

• **Human-centered computing** → HCI theory, concepts and models; Computer supported cooperative work.

Keywords

human-AI teaming, AI adoption, behavioral ontology, delegation, calibration, collaboration health, adoption diagnostics

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1 Position: A Dynamic Relationship the Teaming Frame Only Half-Covers

The workshop asks what work the teaming construct actually does, and for whom. From where I sit—an industrial research lab that both studies enterprise AI adoption and is asked to improve it—my position is this: **human-AI collaboration is a dynamic relationship with observable trajectories, and the teaming frame—built to model collective, reciprocal team states—is the wrong unit of analysis for diagnosing them, which are individual and non-reciprocal.**

This is not the claim that team science is static: it is not. Tuckman and Jensen [29] added *adjourning* to cover a group’s dissolution, and a substantial dynamics-of-team-cognition literature now models how team knowledge emerges, learns, and re-synchronizes over time [10, 11, 32]. The claim is narrower and harder to wave away: that literature theorizes the emergence of *collective, reciprocal* states—shared mental models, transactive memory, mutual monitoring—among members who each hold a stake in the outcome. Two of the trajectories that decide enterprise AI deployments are neither collective nor reciprocal. Whether a worker is *deepening* their practice is an individual expertise trajectory; whether they are quietly *disengaging* is individual technology discontinuance. Each is addressed to a partner that does not reciprocate. Even Grand et al. [11], in the foundational dynamic account, concede that the domain remains largely static in how it articulates these processes. Plateau and disengagement are not edge cases in enterprise AI—they are the *modal* outcomes, and the vocabulary for them lives outside the teaming tradition.

This is easy to miss because industry has taken the metaphor literally, and the literal version hides the gap. Seeber et al. [27] set a research agenda around “machines as teammates”; Microsoft’s Work Trend Index now describes firms organized as hybrid human-agent teams in which employees manage agents [19]; vendors sell “digital workers” that are onboarded, credentialed, and evaluated; analysts supply maturity models staging progress toward the arrangement [1, 8, 33]. The frame is not empty—it licenses organizational design and rightly shifts the unit of analysis from tool to work system. But teammate status here is *stipulated*—a tool becomes a teammate by being described as one, in a deck—and the metaphor quietly imports reciprocity—backup behavior, mutual monitoring, shared accountability [26]—that Klein et al. [14] warned two decades ago the deployed reality does not supply. What it supplies is a *supervisory* relation: one party sets goals, allocates work, evaluates returns, and keeps accountability.

That mismatch has a methodological consequence. If the teaming frame presupposes a functioning team, it presupposes something prior: **a human whose own practice with AI is sound.** Before

Table 1: The four behavioral axes of the HAI Behavioral Lens. An axis defines a *family* of interaction patterns sharing a diagnostic question; each position along it is an independently observable pattern with its own telemetry signature and intervention implications. The rightmost column gives the nearest construct in the teaming literature—two axes have none.

Axis	Diagnostic question	Patterns (positions on the axis)	Grounding	Nearest teaming construct
Delegation	Is the handoff appropriate to the stakes, and is the return evaluated?	over-delegation · calibrated · skill-protective · declining	Trust and reliance in automation [16, 23]	Task allocation; monitoring <i>minus</i> mutuality
Steering	Is the person effectively directing the AI toward their goal?	frustrated retry · productive iteration · authorship reassertion	Self-regulated learning [24, 34]	Closed-loop communication; common-ground repair
Abandonment	Is the person pulling away, and what is driving it?	post-error abandonment · declining engagement · stable · re-engagement	Algorithm aversion; discontinuance [5, 7]	none
Deepening	Is the practice becoming more sophisticated, or has it plateaued?	stagnation · experimentation · progressive deepening · mastery	Expertise development; practice lens [6, 21]	none

asking whether a person and an AI *are* a team, we have to be able to ask whether the person is working well with AI *at all*—and answer it behaviorally, from interaction traces, rather than attributing the relationship’s quality to the agent. The argument runs in three moves: that relationship is observable along four axes, two of which are precisely where teaming vocabulary runs out (§4); read together over time they yield not a score but a collaboration health profile (§5); only then can we ask what teaming adds on top (§6).

2 What Team Science Lends Us: Dynamism and Health Profiles

Before turning to the vacuum, it is worth naming what the human-team analogy genuinely earns—because it is narrower, and more useful, than the teammate rhetoric assumes. When two people join a team, nobody expects the relationship to be static. Tuckman’s stage model describes teams evolving through phases with distinct behavioral signatures that observers can recognize and leaders can act on [28]. Salas et al. [25] established that team effectiveness is *diagnosable* through observable indicators—communication, coordination, trust calibration, shared mental models developing over time. Practitioner instruments follow the same logic: McKinsey’s Team Effectiveness Index reports that measurable team-health drivers explain most of the variance between low- and high-performing teams [18].

The transferable insight is not “AI is a teammate.” It is two properties of how team science *measures*: collaboration health is **dynamic**—it develops, plateaus, regresses, and responds to targeted intervention—and it is read as a **profile** of indicators, never a single number. Human-AI collaboration shares both properties, and this is what the analogy genuinely earns. But team science supplies them at the *collective* level only. It does not supply the two individual, non-reciprocal trajectories that decide enterprise AI use, whose vocabulary lives instead in expertise development [6] and technology discontinuance [7, 30]. I return to that asymmetry in §6.

3 The Vacuum Underneath: Quantity Without Quality

Between the teammate rhetoric and the daily reality sits a measurement vacuum. What enterprises actually hold is seats provisioned, messages per worker, weekly active users, feature breadth, and a maturity stage [12, 15, 20]. All of it is volume—the equivalent of assessing a human team’s health by checking whether members clock in. A marketing department logging 2,400 sessions and an engineering department logging 8,100 are diagnostically equivalent under every platform in production, though one may be integrating AI into complex work while the other performs mandated use without ever routing output into a deliverable.

This matters because the evidence on whether the combination works is not reassuring. Vaccaro et al. [31], meta-analyzing 106 experiments, found human-AI combinations performing on average *worse* than the better of human alone or AI alone, with gains concentrated in particular task types. Industry reporting converges: most enterprises report limited returns from the majority of their AI initiatives [3, 4]. If combinations routinely underperform their components, the interesting variance lives in *how the person works with the system*—precisely the quantity nothing measures. And the failure modes are not benign: cognitive offloading that degrades analytical capacity [9], skill atrophy from sustained over-delegation, and miscalibrated reliance [22] all develop invisibly while usage metrics look healthy.

A few efforts are moving into the middle. Anthropic’s AI Fluency Index classifies behaviors in anonymized conversation logs against a four-dimension framework, and reports that *diligence* has no observable signature in chat logs at all [2]—the dimension that matters most for accountability is the one telemetry cannot see. We have an aspirational construct at the top of the stack, volume metrics at the bottom, and no vocabulary in between.

4 Four Behavioral Axes

Rather than asking “is this a team?”, ask “what is this person’s practice, and is it getting better?” We arrived at four axes by starting from an observability constraint: given interaction telemetry, which facets of a worker’s relationship with an AI tool are actually visible? A subsequent literature review found each axis converging on

an established research tradition, which we take as corroboration rather than derivation (Table 1).

Three properties matter for this workshop. **First, these are axes, not archetypes.** An axis defines a family of interaction patterns that share a diagnostic question, and each position along it is an independently classifiable pattern with its own telemetry signature. A person is not “a frustrated retriever”; they are currently exhibiting the frustrated-retry pattern on the steering axis, and the diagnostic question is what holds them there. Positions, not personas.

Second, the axes are not structurally parallel. Delegation is assessable cross-sectionally; steering is a within-session trajectory; abandonment and deepening are cross-session trajectories that no single session can reveal. Half the framework is diachronic—and that half is individual and non-reciprocal, which is exactly why team cognition, built on collective emergent states, supplies no construct for it.

Third, the axes are non-orthogonal, and the interactions are where diagnosis lives. Steering quality gates deepening; persistent frustrated retry at one complexity level prevents progression, so a stagnating user is often a steering problem misread as a motivation problem. Declining delegation scope precedes abandonment, often by weeks, which makes it an early-warning signal rather than a symptom. Plateau—stable engagement with no deepening—is a distinct third state, and plausibly the modal long-run state of enterprise AI use.

The payoff is that diagnosis couples to action. “Marketing has low AI usage, therefore...” becomes “marketing shows frustrated retry, therefore targeted prompt-design training on marketing use cases.” Same telemetry, different vocabulary, an intervention at the end of it.

5 From Axes to a Collaboration Health Profile

The insight team science lends us—that health is a *profile*, not a score—is what turns four axes into a diagnostic. A sports-medicine physician does not declare an athlete healthy on resting heart rate alone; they assemble a profile across cardiovascular, musculoskeletal, neurological, and metabolic indicators, where the *pattern* across indicators and the *trajectory* over time carry the diagnostic content.

The HAI Behavioral Lens generates an analogous **collaboration health profile**, read on three dimensions. **Composition:** which patterns are present, and what is their valence—productive, miscalibrated, or neutral? **Coherence:** do co-occurring patterns reinforce one another or create a diagnostic tension that needs resolving? **Trajectory:** which direction is each axis moving over weeks and months? A person exhibiting calibrated delegation alongside progressive deepening is on a healthy trajectory. A person showing over-delegation alongside declining verification effort is developing a dependency pattern that demands intervention—even if their usage metrics look exemplary.

I call the behavioral practices that sustain healthy human-AI collaboration **HAIGiene**, by deliberate analogy with preventive health. Just as physical hygiene creates conditions for health without guaranteeing it, HAIGiene names the behavioral substrate—active verification, calibrated delegation, professional-voice preservation, expanding skill integration—from which productive outcomes are more likely to emerge. And just as preventive medicine detects

warning signs before symptoms turn acute, the lens treats declining delegation and frustrated retry as early-warning signals to be acted on before they cascade into disengagement or entrenched dependency. This is what a *diagnostic* account looks like, as against the *organizational* one.

6 What This Gives Teaming, and What Teaming Cannot Give Back

Two axes map onto teaming vocabulary productively. Delegation is task allocation and interdependence, but as a *per-prompt behavioral variable* rather than an ex-ante design parameter—a genuine contribution back to a literature where allocation is usually designed in advance and then assumed. Steering is closed-loop communication, adaptability, and common-ground repair; this is where teaming vocabulary earns its keep even in a dyad. Verification, the evaluative stage of the delegation loop, is mutual performance monitoring with the mutuality removed, which is to say supervision.

Abandonment and deepening map onto nothing—and it is worth being precise about why, because team science is not as static as a quick reading suggests. Tuckman and Jensen’s *adjourning* stage [29] and the dynamic team-cognition literature [10, 11, 32] do model teams changing over time. But they model *collective* change among *reciprocating* members: a team completing its task and disbanding, or a team’s shared mental model re-synchronizing after a perturbation. Deepening—one worker’s expertise trajectory—and abandonment—one worker’s unilateral discontinuance of a non-reciprocal tool—are neither collective nor reciprocal. Their vocabulary lives in expertise development [6], in technology discontinuance [7], and in Tyre and Orlikowski [30]’s finding that adaptation concentrates in an early window and then routinizes, which is effectively a theory of plateau. **The two axes that most often decide whether a deployment succeeds are the two the teaming frame was not built to hold.**

One further reason the sequence matters concerns aggregation. Suppose A delegates a task, does not verify the output, and passes the result to B. B assumes it was checked, because in a human team mutual monitoring makes that assumption safe [26]. The unverified content becomes load-bearing and no one holds the information needed to know it. A team of calibrated delegators is a team; a team of over-delegators is a mechanism for laundering unverified output into shared work product, and teaming structures layered on top distribute that miscalibration rather than correcting it.

7 What I Want to Argue About

- (1) Is individual practice genuinely *prior* to teaming? Concretely: name a teaming intervention that works on a population of over-delegators.
- (2) Should teaming research import the vocabulary of discontinuance and expertise development to cover trajectory, or does that absence indicate teaming is the wrong frame for enterprise AI work?
- (3) Who holds the profile and at what granularity? Axis positions are claims about a person’s interaction behavior: held by the individual this is personal informatics [17], held by the employer it is performance surveillance [13].

Human-AI collaboration is a dynamic relationship, and the teaming frame gives us language for only half of it—the half that is collective and reciprocal. The other half is the individual trajectory of a worker deepening or disengaging, and covering that gap requires building the missing diagnostic instrument.

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